

SNOMED CT Terminology Server

External Installation Test Guide

IRIS Terminal tests and MCP client prompts for validating a deployed SNOMED CT Server

Use this guide after installing the SNOMED CT Terminology Server to confirm concept lookup, lexical search, hierarchy navigation, subsumption checks, semantic search, and MCP access are working correctly.

Result variability

Exact JSON ordering and available descendants can vary by SNOMED CT edition and release. A test passes when the response is well-formed, clinically plausible, and includes the highlighted expected concept or relationship.

Quick start

Open an IRIS terminal session inside the running SNOMED container:

```
docker exec -it snomed-iris iris session IRIS -U USER
```

Semantic search requires Ollama to be running and the embedding model to be available:

```
docker exec ollama ollama list  
# Expected: all-minilm appears in the model list
```

Pass criteria

- Concept lookup returns JSON that includes the requested ConceptId, Active=1, an FSN, and description terms.
- Lexical and semantic searches return JSON arrays of matching concepts.
- Hierarchy calls return arrays of concept IDs for parents, children, ancestors, or descendants.
- Subsumption checks return 1 for true IS-A relationships and 0 for false relationships.
- MCP tools are discoverable in the client and can execute the same categories of tests.

1. IRIS Terminal Test Suite

Run the following ObjectScript commands at the IRIS terminal prompt. The examples assume the container is named **snomed-iris** and the USER namespace contains the SNOMED server classes and data.

1.1 Concept lookup

```
Write ##class(SNOMED.Tools.Lookup).GetConcept("73211009")
Write ##class(SNOMED.Tools.Lookup).GetConcept("22298006")
```

Concept ID	Expected confirmation
73211009	Returns Diabetes mellitus (disorder), with descriptions such as Diabetes mellitus and DM - Diabetes mellitus.
22298006	Returns Myocardial infarction (disorder), with descriptions such as Heart attack and Myocardial infarction.

1.2 Lexical search

```
Write ##class(SNOMED.Search).Lexical("myocardial infarction", 5)
Write ##class(SNOMED.Search).Lexical("atrial fibrillation", 5)
Write ##class(SNOMED.Search).Lexical("heart failure", 5)
```

Search term	Expected first-page results
myocardial infarction	Includes ConceptId 22298006 with term Myocardial infarction.
atrial fibrillation	Includes ConceptId 49436004 with term Atrial fibrillation.
heart failure	Includes ConceptId 84114007 with term Heart failure.

1.3 Hierarchy navigation

```
Write ##class(SNOMED.Hierarchy).GetParents("44054006")
Write ##class(SNOMED.Hierarchy).GetChildren("73211009")
Write ##class(SNOMED.Hierarchy).GetAncestors("44054006")
Write ##class(SNOMED.Hierarchy).GetDescendants("73211009")
```

Function	Expected confirmation
GetParents(44054006)	Returns ["73211009"], confirming Type 2 diabetes mellitus has Diabetes mellitus as a direct parent.
GetChildren(73211009)	Returns a JSON array of direct child concepts, including 44054006 in the US Edition sample data.
GetAncestors(44054006)	Returns a JSON array including 73211009, 404684003, and 138875005.
GetDescendants(73211009)	Returns a non-empty JSON array of diabetes-related descendant concept IDs; output may be long.

1.4 Subsumption / IS-A checks

```
Write ##class(SNOMED.Hierarchy).IsDescendantOf("44054006", "73211009")
Write ##class(SNOMED.Hierarchy).IsDescendantOf("71620000", "73211009")
```

Test	Expected result	Meaning
44054006 IS-A 73211009	1	Type 2 diabetes mellitus is a subtype of Diabetes mellitus.
71620000 IS-A 73211009	0	Fracture of femur is not a subtype of Diabetes mellitus.

1.5 Semantic search

Semantic search requires Ollama and the same embedding model used during data generation. When IRIS runs in Docker and Ollama runs on the host or another container, validate the URL before testing.

```
Write ##class(SNOMED.Search).Semantic("chest pain when breathing", 5, 0.5, "http://host.docker.internal:11434")
Write ##class(SNOMED.Search).Semantic("difficulty swallowing food", 5, 0.5, "http://host.docker.internal:11434")
Write ##class(SNOMED.Search).Semantic("rash on both arms", 5, 0.5, "http://host.docker.internal:11434")
```

Query	Expected top or near-top result
chest pain when breathing	Chest pain on breathing (finding), ConceptId 274664007, should appear with a high similarity score.
difficulty swallowing food	Difficulty swallowing food (finding), ConceptId 288945004, should appear with a high similarity score.
rash on both arms	Rash-related disorder or finding results should be returned, such as Erythematous rash, Generalized rash, or similar terms.

Semantic search signature

Parameter	Type	Default	Description
query	String	required	Natural language description of the clinical concept to find.
maxResults	Integer	10	Maximum number of concepts to return.
minScore	Numeric	0.5	Minimum cosine similarity threshold from 0 to 1.
ollamaUrl	String	http://localhost:11434	Ollama server URL for query embeddings.
modelName	String	all-minilm	Embedding model name; must match the model used during generation.

2. MCP Server Test Suite

These tests validate the SNOMED tools exposed through the Model Context Protocol. Use them after the IRIS terminal tests pass.

2.1 Configure an MCP-compatible client

For Claude Code, place a configuration like this in the project **.mcp.json**. This form runs **iris-mcp-server** inside the SNOMED IRIS container, so no local macOS or Windows binary is required.

```
{
  "mcpServers": {
    "snomed": {
      "command": "docker",
      "args": [
        "exec", "-i", "snomed-iris",
        "/usr/irissys/bin/iris-mcp-server",
        "--config", "/opt/iris/config/mcp-config.toml",
        "run"
      ]
    }
  }
}
```

Activate the configuration

Restart the MCP client after saving the configuration. When prompted, approve the SNOMED MCP server. In Claude Code, use **/mcp** to confirm the server is connected and tools are available.

2.2 MCP tool discovery

A working server should expose the SNOMED lookup, search, hierarchy, and subsumption tools. The source Word document references 9 SNOMED tools, including lookup, lexical search, semantic search, parent/child/ancestor/descendant navigation, and IS-A checking.

Category	User prompt	Expected behavior
Basic lookup	What is SNOMED concept 73211009?	Client calls the SNOMED concept lookup tool and returns Diabetes mellitus.
Lexical search	Search SNOMED for concepts related to myocardial infarction	Client calls lexical search and returns myocardial infarction concepts.
Hierarchy	What are the parent concepts of Type 2 diabetes mellitus (44054006)?	Client returns Diabetes mellitus (73211009).
Subsumption	Is Type 2 diabetes mellitus (44054006) a subtype of Diabetes mellitus (73211009)?	Client calls IS-A/subsumption and returns true.
Semantic search	Find SNOMED concepts semantically similar to chest pain when breathing	Client calls semantic search and returns chest-pain-related findings.

2.3 Recommended MCP prompt checklist

#	Prompt	Purpose
1	What is SNOMED concept 73211009?	Basic concept lookup.
2	Look up concept 22298006	Verify myocardial infarction lookup.
3	Search SNOMED for concepts related to myocardial infarction	Lexical keyword search.
4	Find SNOMED terms matching atrial fibrillation	Lexical keyword search.
5	Search for concepts about type 2 diabetes	Lexical search with diabetes terms.
6	What are the parent concepts of Type 2 diabetes mellitus (44054006)?	Parent navigation.
7	Show me the children of Diabetes mellitus (73211009)	Child navigation.
8	What are all the ancestors of concept 44054006?	Ancestor navigation.
9	Get the first 20 descendants of Clinical finding (404684003)	Descendant navigation.
10	Is Type 2 diabetes mellitus (44054006) a subtype of Diabetes mellitus (73211009)?	Positive IS-A test.
11	Is fracture of femur (71620000) a type of diabetes?	Negative IS-A test.
12	Find SNOMED concepts semantically similar to chest pain when breathing	Semantic search.
13	What concepts match the meaning of difficulty swallowing food?	Semantic search.
14	Find concepts related to rash appearing on both arms	Semantic search.
15	I have a patient with chest pain that worsens on inspiration. What SNOMED concepts might apply? Check if any are subtypes of Clinical finding.	Multi-step lookup plus hierarchy reasoning.
16	Find all types of diabetes in SNOMED and tell me which ones are subtypes of Diabetes mellitus.	Multi-step search plus subsumption.

2.4 Example multi-tool MCP validation

Prompt:

Is 73211009 a type of metabolic disease?

Expected sequence:

1. LexicalSearch(term: "metabolic disease", maxResults: 3)
Expected: finds Metabolic disease, ConceptId 75934005.
2. IsA(conceptId: "73211009", ancestorId: "75934005")
Expected: {"isDescendant":1,"conceptId":"73211009","ancestorId":"75934005"}
3. Final answer: Diabetes mellitus (73211009) is a type of Metabolic disease (75934005).

3. Troubleshooting

Symptom	Checks and fixes
IRIS terminal cannot connect	Confirm the container is running: <code>docker ps grep snomed-iris</code> . Check that the namespace is <code>USER</code> and that the image was started with the expected ports.
Concept lookup returns empty or errors	Confirm SNOMED RF2 data was loaded, tables contain records, and the active edition matches the expected <code>ConceptIds</code> .
Semantic search fails	Confirm Ollama is running, <code>all-minilm</code> is pulled, <code>OLLAMA_URL</code> is reachable from the IRIS container, and embeddings have been generated.
MCP tools do not appear	Restart the client, approve the MCP server, verify <code>.mcp.json</code> syntax, and run <code>/mcp</code> or the client equivalent to reconnect.
MCP discovery returns HTTP 403	The source notes that the effective web user needs <code>READ</code> access to <code>%DB_USER</code> . Confirm installer security setup and roles for the web application users.
MCP execution returns SQL prepare errors	The source notes that the execution context needs <code>SELECT</code> privilege on the <code>SNOMED</code> schema. Confirm SQL privileges were granted during installation.

Appendix: Minimal smoke-test script

When you want a fast confidence check, run these commands first. They validate the four core paths: lookup, lexical search, hierarchy, and subsumption.

```
docker exec -it snomed-iris iris session IRIS -U USER

Write ##class(SNOMED.Tools.Lookup).GetConcept("73211009"),!
Write ##class(SNOMED.Search).Lexical("heart failure", 5),!
Write ##class(SNOMED.Hierarchy).GetParents("44054006"),!
Write ##class(SNOMED.Hierarchy).IsDescendantOf("44054006", "73211009"),!
```

Expected smoke-test outcome: the lookup returns Diabetes mellitus, lexical search returns Heart failure concepts, parents returns ["73211009"], and the IS-A check returns 1.